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标题	Late‐Stage Modification of Halotryptophan‐Containing Peptides Via Negishi Cross‐Coupling
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摘要（原文节选）

ABSTRACTLate‐stage functionalization of peptides provides powerful opportunities for the rapid diversification of bioactive sequences. Among the available strategies, Negishi cross‐coupling represents a versatile and orthogonal method for the installation of alkyl substituents under mild conditions. We describe the applicability of Negishi cross‐coupling to increasingly complex bromotryptophan‐containing peptides. While conditions effective for bromo‐l‐tryptophan proved unsuitable for peptide substrates, leading to ⋯

数据表 · TABLE 1（4 行，含数值行 4 行）

Scope of alkyl iodides in the Negishi cross‐coupling of Ac‐Ala‐Trp(6‐Br)‐Gly‐NH2 (Scheme 2, see also Supporting Information).

Entry	Alkyl iodide	#	Conversion (%) a , b	Yield (%)
1		1a	79 (61,18)	59
2		1b	89 (65, 24)	55
3		1c	>99 (99, 0)	97
4		1d	92 (87, 5)	86

脚注（逐字）：a Determined by UHPLC‐MS at 220 nm. b Values in brackets indicate the percentage of product and debrominated peptide (relative to conversion).

正文对应句（逐字）：A stock solution of the peptide in anhydrous DMF was prepared and purged with argon prior to addition to a reaction vial containing Pd(amphos)2Cl2, activated zinc dust, and the alkyl iodide dissolved in DMF and likewise purged with argon. The suspension was vigorously stirred at 40 ° C, and reaction p

数据表 · TABLE 2（4 行，含数值行 4 行）

Negishi cross‐coupling of 7‐bromo‐l‐tryptophan containing pentapeptides with N α ‐Boc‐l‐3‐iodoalanine methyl ester in DMF to investigate the influence of different amino acids' side chains (Scheme 3, see also Supporting Information).

Entry	Peptide	#	Solvent	Conversion (%) a , b	Yield
5	Ac‐Leu‐Val‐Trp(7‐Br)‐Ala‐Gly‐NH2	6a	DMF	18 (9,9)	22 c
6	Ac‐Asp‐Val‐Trp(7‐Br)‐Ala‐Gly‐NH2	7a	DMF	13 (0,13)	n.d. d
7	Ac‐Ser‐Val‐Trp(7‐Br)‐Ala‐Gly‐NH2	8a	DMF	58 (32, 26)	21
8	Ac‐Met‐Val‐Trp(7‐Br)‐Ala‐Gly‐NH2	9a	DMF	82 (25, 57)	29 c

脚注（逐字）：a Determined by UHPLC‐MS at 220 nm. b Values in brackets indicate the percentage of product and debrominated peptide (relative to conversion). c Contamination with side‐product Boc‐Ala‐OMe. d Not determined.

正文对应句（逐字）：A stock solution of the peptide in anhydrous DMF was prepared and purged with argon prior to addition to a reaction vial containing Pd(amphos)2Cl2, activated zinc dust, and the alkyl iodide dissolved in DMF and likewise purged with argon. The suspension was vigorously stirred at 40 ° C, and reaction p

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